

## KEATON BEACH, FL, USA

WMO: 994650

Lat: 29.8187N

Lon: 83.5935W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 0.2        | 2.2 | -9.3                          | 1.7 | 3.1  | -6.2 | 2.2 | 5.6  | 12.6                  | 14.6 | 11.2 | 15.9 | 2.8                   | 30   | 0.425 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 5.5                    | 31.0            | 25.0 | 30.4 | 25.1 | 30.0 | 25.0 | 26.9                | 29.7 | 26.4 | 29.4 | 26.0 | 29.0 | 4.3                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.9                            | 21.3 | 29.2 | 25.5 | 20.7 | 28.8 | 25.1 | 20.2 | 28.5 | 84.0          | 29.7 | 82.0 | 29.5 | 80.2 | 29.1 | 28.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 11.2              | 9.4  | 8.3 | -3.4                       | 33.0 | 1.9                | 1.1 | -4.7                                               | 33.7 | -5.8       | 34.3 | -6.9       | 34.9 | -8.3       | 35.7 |    |    |
|                   |      |     | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 19.7   | 11.3  | 13.0  | 15.6  | 18.9  | 22.7  | 25.8  | 26.6  | 26.6  | 25.5  | 21.5  | 15.9  | 13.2 |
|                                                                                  | DBStd       | 6.37   | 4.75  | 4.14  | 3.60  | 2.73  | 2.22  | 1.39  | 1.17  | 1.34  | 1.62  | 3.26  | 4.07  | 4.49 |
|                                                                                  | HDD10.0     | 91     | 43    | 17    | 5     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 21   |
|                                                                                  | HDD18.3     | 765    | 220   | 153   | 94    | 26    | 2     | 0     | 0     | 0     | 0     | 12    | 92    | 167  |
|                                                                                  | CDD10.0     | 3647   | 83    | 101   | 179   | 266   | 394   | 473   | 514   | 515   | 464   | 356   | 183   | 119  |
|                                                                                  | CDD18.3     | 1281   | 2     | 3     | 11    | 42    | 138   | 223   | 256   | 257   | 214   | 109   | 20    | 7    |
|                                                                                  | CDH23.3     | 11233  | 1     | 2     | 4     | 79    | 919   | 2231  | 2793  | 2703  | 1829  | 606   | 64    | 3    |
| CDH26.7                                                                          | 2971        | 0      | 0     | 0     | 1     | 75    | 553   | 904   | 884   | 488   | 66    | 1     | 0     |      |
| Wind                                                                             | WSAvg       | 3.9    | 3.7   | 3.9   | 4.5   | 4.4   | 4.5   | 4.3   | 3.8   | 3.7   | 3.6   | 3.4   | 3.5   | 3.8  |
| Precipitation                                                                    | PrecAvg     | 1379   | 106   | 83    | 109   | 79    | 58    | 165   | 213   | 208   | 144   | 83    | 54    | 74   |
|                                                                                  | PrecMax     | 1809   | 244   | 300   | 380   | 219   | 136   | 409   | 490   | 421   | 410   | 208   | 121   | 200  |
|                                                                                  | PrecMin     | 911    | 30    | 14    | 13    | 10    | 2     | 55    | 106   | 107   | 49    | 10    | 2     | 15   |
|                                                                                  | PrecStd     | 240    | 55    | 60    | 78    | 53    | 39    | 62    | 82    | 81    | 91    | 52    | 32    | 50   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 22.2  | 23.5  | 24.0  | 25.8  | 29.6  | 31.3  | 31.9  | 31.8  | 31.8  | 29.6  | 26.6  | 23.7 |
|                                                                                  |             | MCWB   | 18.9  | 20.2  | 17.7  | 21.0  | 22.8  | 24.4  | 24.8  | 25.4  | 23.7  | 23.4  | 21.2  | 18.4 |
|                                                                                  | 2%          | DB     | 20.6  | 21.6  | 22.8  | 25.0  | 28.3  | 30.1  | 30.8  | 30.8  | 30.7  | 28.3  | 25.3  | 22.3 |
|                                                                                  |             | MCWB   | 18.3  | 19.5  | 19.1  | 21.3  | 22.8  | 24.8  | 25.7  | 25.6  | 23.8  | 23.6  | 21.9  | 18.3 |
|                                                                                  | 5%          | DB     | 19.3  | 20.0  | 21.8  | 24.3  | 27.5  | 29.6  | 30.2  | 30.3  | 29.8  | 27.4  | 23.7  | 21.2 |
|                                                                                  |             | MCWB   | 17.6  | 18.0  | 18.9  | 21.0  | 23.0  | 24.9  | 25.4  | 25.5  | 24.1  | 23.2  | 20.1  | 18.1 |
|                                                                                  | 10%         | DB     | 18.0  | 18.8  | 21.0  | 23.6  | 26.8  | 29.1  | 29.7  | 29.9  | 29.0  | 26.4  | 22.2  | 20.1 |
|                                                                                  |             | MCWB   | 16.4  | 17.4  | 18.4  | 20.3  | 22.7  | 24.8  | 25.2  | 25.4  | 24.1  | 22.8  | 19.0  | 18.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 20.2  | 21.3  | 21.8  | 23.6  | 25.5  | 27.2  | 27.4  | 27.6  | 26.8  | 25.7  | 24.4  | 20.4 |
|                                                                                  |             | MCDB   | 21.2  | 22.4  | 22.3  | 25.0  | 28.0  | 29.3  | 30.4  | 30.3  | 29.5  | 27.5  | 25.7  | 20.9 |
|                                                                                  | 2%          | WB     | 19.0  | 20.2  | 20.6  | 22.8  | 24.6  | 26.6  | 26.7  | 26.9  | 25.9  | 25.0  | 22.7  | 19.3 |
|                                                                                  |             | MCDB   | 20.1  | 21.4  | 21.6  | 24.5  | 26.9  | 29.1  | 29.7  | 29.8  | 28.7  | 27.0  | 24.0  | 20.1 |
|                                                                                  | 5%          | WB     | 18.0  | 19.0  | 19.9  | 21.9  | 24.0  | 25.9  | 26.2  | 26.4  | 25.3  | 24.2  | 20.9  | 18.6 |
|                                                                                  |             | MCDB   | 19.0  | 19.7  | 21.1  | 23.6  | 26.3  | 28.7  | 29.3  | 29.5  | 28.4  | 26.5  | 22.5  | 19.6 |
|                                                                                  | 10%         | WB     | 16.9  | 17.9  | 19.1  | 21.3  | 23.6  | 25.4  | 25.7  | 26.0  | 24.9  | 23.5  | 19.8  | 17.8 |
|                                                                                  |             | MCDB   | 17.8  | 18.6  | 20.5  | 23.0  | 25.9  | 28.3  | 29.0  | 29.1  | 28.0  | 25.8  | 21.6  | 19.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.7   | 7.2   | 7.3   | 7.0   | 6.4   | 5.5   | 5.5   | 5.5   | 6.6   | 7.7   | 8.8   | 8.1  |
|                                                                                  |             | MCDBR  | 7.3   | 7.0   | 7.4   | 7.7   | 7.3   | 6.5   | 6.4   | 6.3   | 7.9   | 7.5   | 8.3   | 7.5  |
|                                                                                  |             | MCWBR  | 5.3   | 5.1   | 4.7   | 4.9   | 4.2   | 3.4   | 3.2   | 3.1   | 3.1   | 3.6   | 5.3   | 5.4  |
|                                                                                  | 5% WB       | MCDBR  | 5.6   | 5.0   | 4.9   | 5.6   | 5.2   | 4.9   | 5.1   | 5.1   | 5.8   | 5.8   | 6.6   | 6.6  |
|                                                                                  |             | MCWBR  | 5.1   | 4.5   | 4.3   | 4.8   | 3.7   | 3.2   | 3.1   | 3.1   | 3.1   | 3.8   | 5.6   | 5.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.341  | 0.346 | 0.364 | 0.389 | 0.430 | 0.467 | 0.497 | 0.480 | 0.434 | 0.384 | 0.354 | 0.339 |      |
|                                                                                  | taud        | 2.515  | 2.504 | 2.461 | 2.370 | 2.304 | 2.267 | 2.196 | 2.248 | 2.357 | 2.477 | 2.491 | 2.546 |      |
|                                                                                  | Ebn at Noon | 894    | 921   | 923   | 905   | 864   | 828   | 802   | 814   | 842   | 868   | 870   | 879   |      |
|                                                                                  | Edh at Noon | 89     | 98    | 109   | 123   | 132   | 137   | 146   | 138   | 119   | 100   | 91    | 83    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.26   | 3.91  | 5.19  | 6.23  | 6.85  | 6.23  | 5.98  | 5.54  | 5.06  | 4.59  | 3.65  | 2.93  |      |
|                                                                                  | RadStd      | 0.23   | 0.42  | 0.47  | 0.47  | 0.38  | 0.39  | 0.39  | 0.40  | 0.42  | 0.34  | 0.24  | 0.25  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) | Regional (30 neighbors) | +0.47 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | -21         | -78     | +160    | +110    |

Nomenclature: See separate page